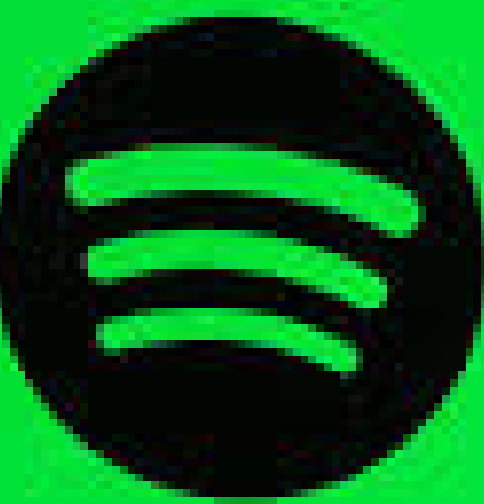


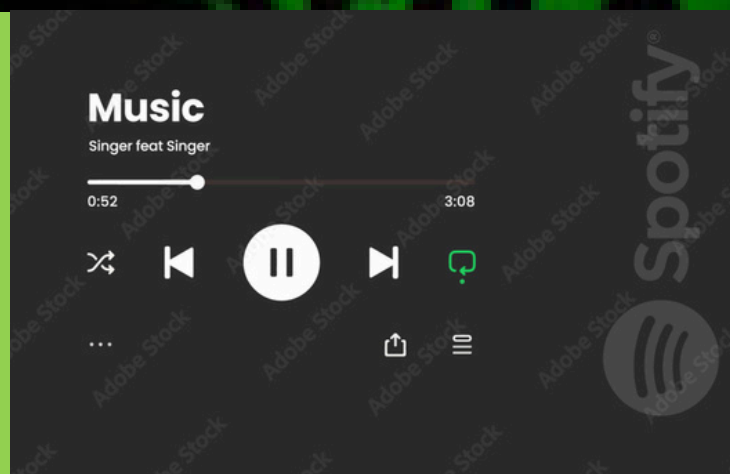


Spotify



Exploratory Data Analysis

Info:-



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Overview

These five CSV files together form a comprehensive Spotify analytics dataset. The tracks dataset contains song-level information along with key audio features such as danceability, energy, tempo, valence, and popularity. The artist dataset focuses on artist-wise metrics, helping analyze popularity and musical characteristics at the artist level. The genre dataset provides genre-based aggregation, useful for understanding genre trends and preferences. The year-wise dataset captures how musical features have evolved over time, enabling trend analysis across different years. Finally, the tracks-with-genres dataset links individual songs with their respective genres, allowing combined analysis of song features and genre popularity. Together, these datasets enable detailed exploratory data analysis, visualization, and insights into music trends and user preferences on Spotify.

OBJECTIVE OF THE PROJECT

- 1. To analyze Spotify music datasets in order to understand overall music trends and listening patterns across different years, genres, and artists.**
- 2. To study important audio features such as danceability, energy, tempo, loudness, and valence, and examine how these features define the characteristics of songs.**
- 3. To identify the key factors that influence song popularity on Spotify by comparing audio features with popularity scores.**
- 4. To analyze genre-wise and artist-wise data to understand user preferences and identify the most popular music categories and artists on the platform.**
- 5. To apply Exploratory Data Analysis (EDA) and visualization techniques such as bar charts, line graphs, scatter plots, and heatmaps to extract meaningful insights and support data-driven decision-making in the music industry.**

TOOLS & TECHNOLOGIES USED

- Pthyan
- Pandas
- Seaborn
- Matplotlib
- Google Collab
- Canva (for presentation)



DATA EXPLORATION

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df_data = pd.read_csv('/content/data.csv')
df_artist = pd.read_csv('/content/data_by_artist.csv')
df_genres = pd.read_csv('/content/data_by_genres.csv')
df_year = pd.read_csv('/content/data_by_year.csv')
df_w_genres = pd.read_csv('/content/data_w_genres.csv')
```

```
dataset = {"data": df_data, "Artist": df_artist,
           "Genres": df_genres, "Year": df_year, "With Genres": df_w_genres}
for name, df in dfs.items():
    print(f"\n{name} Dataset Shape:", df.shape)
    display(df.head())
```

...	valence	year	acousticness	artists	danceability	duration_ms	energy	explicit	id	instrumentalness	key	liveness	loudness	mode	name
0	0.0594	1921	0.982	['Sergei Rachmaninoff', 'James Levine', 'Berli...	0.279	831667	0.211	0	4BJqT0PrAfxzMOxylFOIz	0.878000	10	0.665	-20.096	1	Piano Concerto No. 3 in D Minor, Op. 30: III. ...
1	0.9630	1921	0.732	['Dennis Day']	0.819	180533	0.341	0	7xPhfUan2yNtyFG0cUWk18	0.000000	7	0.160	-12.441	1	Clancy Lowered the Boom
2	0.0394	1921	0.961	['KHP Kridhamardawa Karaton Ngayogyakarta Hadi...	0.328	500062	0.166	0	1o6i8BgIA6yiDMrIElygv1	0.913000	3	0.101	-14.850	1	Gati Bali
3	0.1650	1921	0.967	['Frank Parker']	0.275	210000	0.309	0	3ftBPcC5vPBKxYSee08FDH	0.000028	5	0.381	-9.316	1	Danny Boy
4	0.2530	1921	0.957	['Phil Regan']	0.418	166693	0.193	0	4d6HGyGT8e121BsdKmw9v6	0.000002	3	0.229	-10.096	1	When Irish Eyes Are Smiling

Artist Dataset Shape: (28680, 15)

mode	count	acousticness	artists	danceability	duration_ms	energy	instrumentalness	liveness	loudness	speechiness	tempo	valence	popularity	key
------	-------	--------------	---------	--------------	-------------	--------	------------------	----------	----------	-------------	-------	---------	------------	-----

- The first image shows a preview of the Spotify track-level dataset, which includes detailed audio features such as danceability, energy, valence, acousticness, loudness, tempo, and popularity along with artist and song information. This confirms the dataset is rich and suitable for in-depth music analysis.
- The data spans very early years (e.g., 1921), indicating a wide historical range, which is useful for trend and year-wise analysis.
- The presence of both instrumental and vocal tracks (instrumentalness values) highlights diversity in musical styles.
- The second image confirms that all five CSV files are successfully loaded into Google Colab without errors, validating correct file handling and setup.
- Dataset shapes indicate a large and structured dataset, making it ideal for EDA, visualization, and extracting meaningful insights about Spotify music trends and user preferences

Descriptive Statistics



```
df_data.describe()
```

year	acousticness	danceability	duration_ms	energy	explicit	instrumentalness	key	liveness	loudness	mode	popularity	speechiness	tempo
170653.000000	170653.000000	170653.000000	1.706530e+05	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000
76.787241	0.502115	0.537396	2.309483e+05	0.482389	0.084575	0.167010	5.199844	0.205839	-11.467990	0.706902	31.431794	0.098393	116.861590
25.917853	0.376032	0.176138	1.261184e+05	0.267646	0.278249	0.313475	3.515094	0.174805	5.697943	0.455184	21.826615	0.162740	30.708533
21.000000	0.000000	0.000000	5.108000e+03	0.000000	0.000000	0.000000	0.000000	0.000000	-60.000000	0.000000	0.000000	0.000000	0.000000
56.000000	0.102000	0.415000	1.698270e+05	0.255000	0.000000	0.000000	2.000000	0.098800	-14.615000	0.000000	11.000000	0.034900	93.421000
77.000000	0.516000	0.548000	2.074670e+05	0.471000	0.000000	0.000216	5.000000	0.136000	-10.580000	1.000000	33.000000	0.045000	114.729000
99.000000	0.893000	0.668000	2.624000e+05	0.703000	0.000000	0.102000	8.000000	0.261000	-7.183000	1.000000	48.000000	0.075600	135.537000
20.000000	0.996000	0.988000	5.403500e+06	1.000000	1.000000	1.000000	11.000000	1.000000	3.855000	1.000000	100.000000	0.970000	243.507000

- The dataset has 170,653 songs, covering a wide time range from 1921 to 2020, with an average release year around 1977, indicating a strong presence of older music.
- Danceability (mean ≈ 0.54) and energy (mean ≈ 0.48) suggest that most tracks are moderately upbeat and suitable for general listening.
- Valence (mean ≈ 0.53) shows a balanced emotional tone, meaning songs are neither too sad nor too happy overall.
- Duration averages around 230,000 ms (~3.8 minutes), which aligns with standard song lengths.
- Popularity has a low-to-moderate mean (~31), indicating most tracks are not highly popular, with only a few hits.
- Tempo varies widely (0–243 BPM), reflecting diversity across slow, medium, and fast-paced music.

HANDLING MISSING VALUES

```
▶ for name, df in dataset.items():  
    print(f"\n{name} Missing Values:")  
    print(df.isnull().sum())  
  
df_data.drop_duplicates(inplace=True)
```

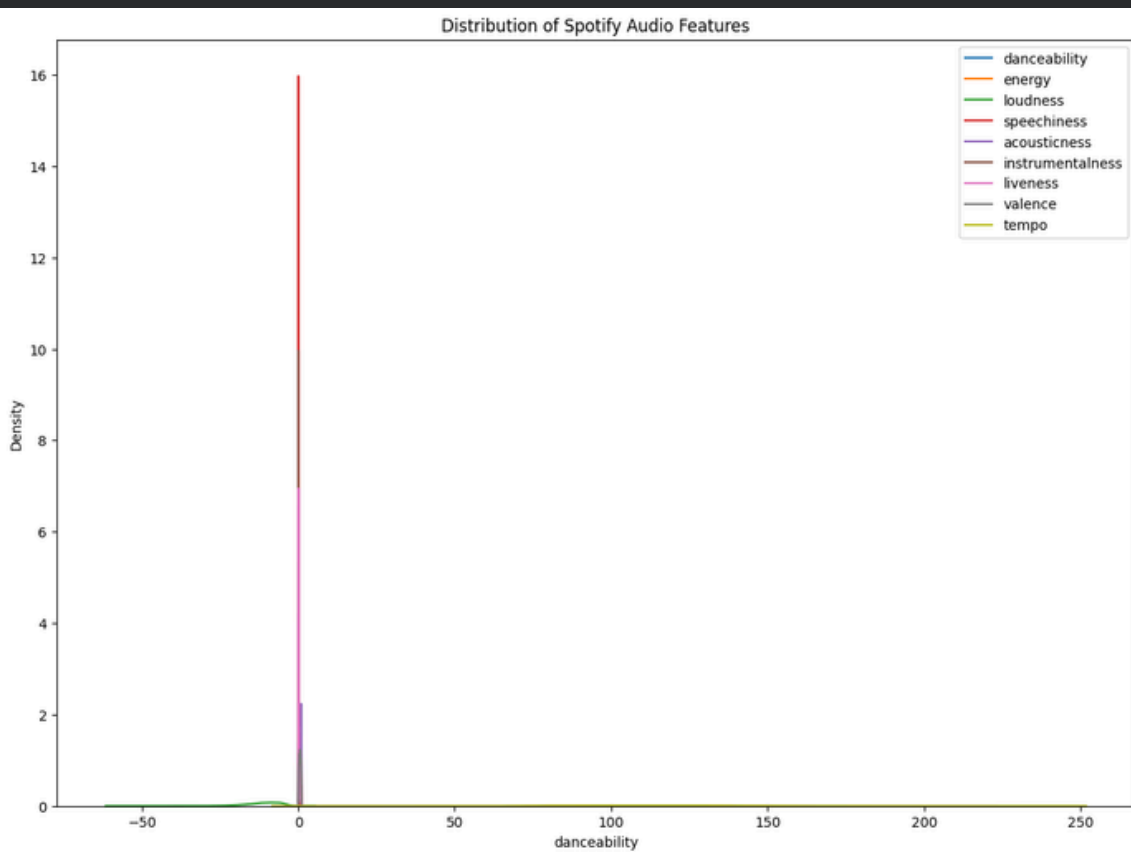
data Missing Values:		Genres Missing Values:		With Genres Missing Values:	
valence	0	mode	0	genres	0
year	0	genres	0	artists	0
acousticness	0	acousticness	0	acousticness	0
artists	0	danceability	0	danceability	0
danceability	0	duration_ms	0	duration_ms	0
duration_ms	0	energy	0	energy	0
energy	0	instrumentalness	0	instrumentalness	0
explicit	0	liveness	0	liveness	0
id	0	loudness	0	loudness	0
instrumentalness	0	speechiness	0	speechiness	0
key	0	tempo	0	tempo	0
liveness	0	valence	0	valence	0
loudness	0	popularity	0	popularity	0
mode	0	key	0	key	0
name	0	dtype: int64		mode	0
popularity	0	Year Missing Values:		count	0
release_date	0	mode	0	dtype: int64	
speechiness	0	year	0		
tempo	0	acousticness	0		
dtype: int64		danceability	0		
Artist Missing Values:		duration_ms	0		
mode	0	energy	0		
count	0	instrumentalness	0		
acousticness	0	liveness	0		
artists	0	loudness	0		
danceability	0	speechiness	0		
duration_ms	0	tempo	0		
energy	0	valence	0		
instrumentalness	0	popularity	0		
liveness	0	key	0		
loudness	0	dtype: int64			
speechiness	0				
tempo	0				
valence	0				
popularity	0				
key	0				
dtype: int64					

- The analysis shows zero missing values across all datasets (Data, Artist, Genres, Year, and With Genres), indicating excellent data completeness.
- Key audio features such as acousticness, danceability, energy, loudness, tempo, and valence are fully populated, making the dataset reliable for analysis and modeling.
- Metadata columns like artists, name, release_date, year, and genres also contain no null values, ensuring consistency in descriptive insights.
- The absence of missing values reduces the need for data imputation or cleaning steps, saving preprocessing time.
- The use of `drop_duplicates()` ensures duplicate records are removed, improving data quality.
- Overall, the dataset is clean, well-structured, and analysis-ready, suitable for EDA, visualization, and machine learning tasks.

Data Visualization



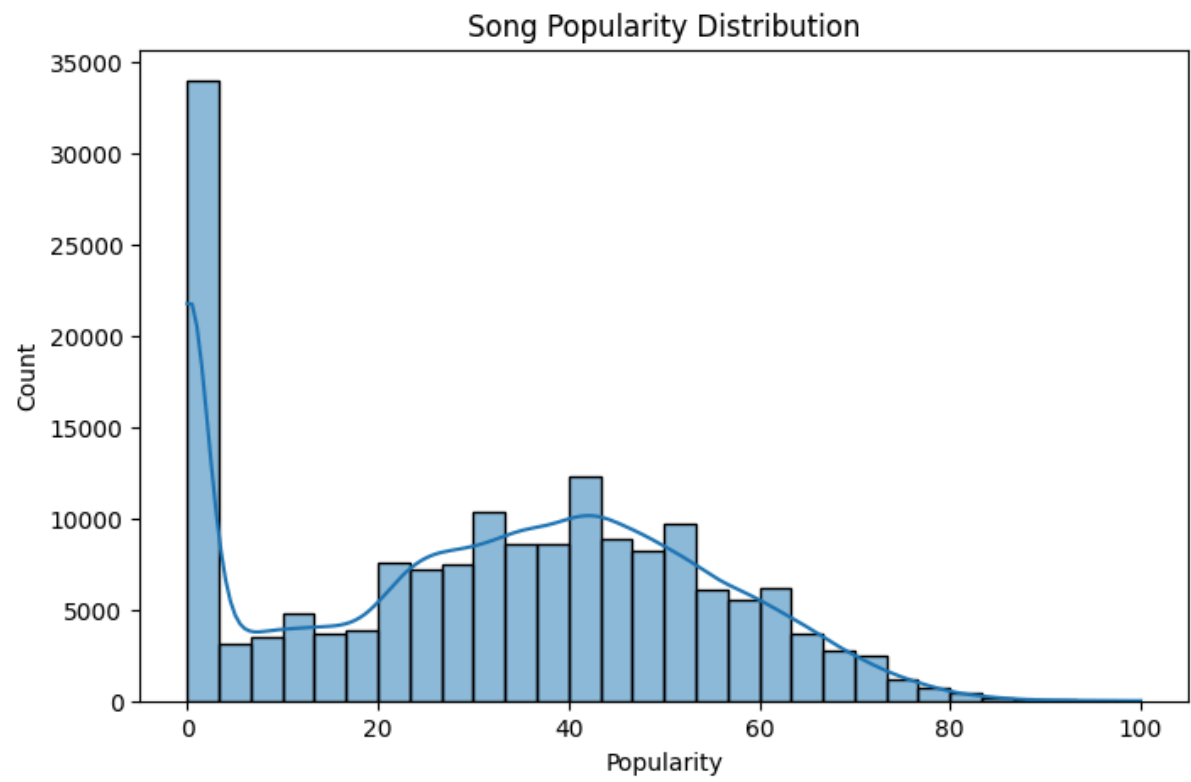
```
audio_features = ['danceability', 'energy', 'loudness', 'speechiness',  
                  'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo']  
plt.figure(figsize=(14,10))  
for feature in audio_features:  
    sns.kdeplot(df_data[feature], label=feature)  
plt.title("Distribution of Spotify Audio Features")  
plt.legend()  
plt.show()
```



- Most audio features are densely clustered near lower values, indicating that the majority of Spotify tracks share similar characteristics in terms of danceability, energy, valence, and acousticness.
- Tempo and loudness show wider spread, reflecting strong variation in song speed and sound intensity across different musical styles.



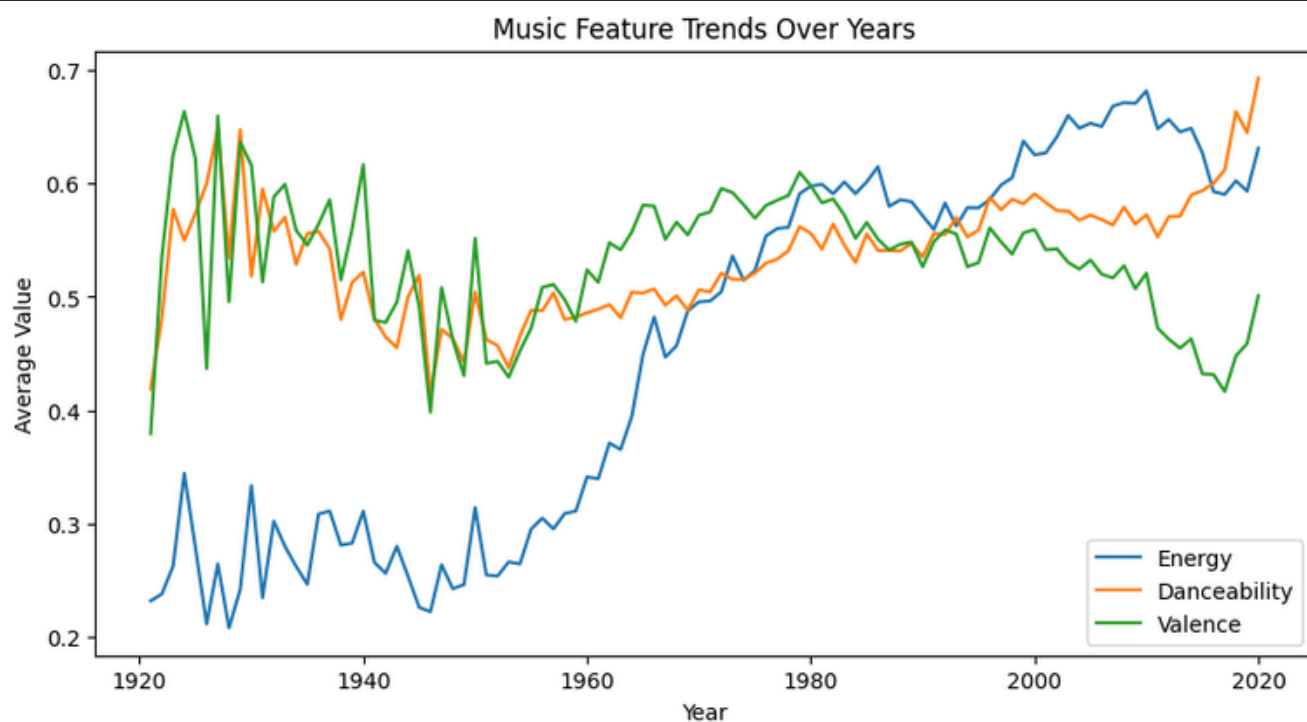
```
plt.figure(figsize=(8,5))  
sns.histplot(df_data['popularity'], bins=30, kde=True)  
plt.title("Song Popularity Distribution")  
plt.xlabel("Popularity")  
plt.ylabel("Count")  
plt.show()
```



1. The graph shows that most songs are concentrated in the low to medium popularity range (0–50), indicating a large number of average-performing tracks.
2. Songs with high popularity (70–100) are very few, highlighting the rarity of major hit songs.
3. The distribution is right-skewed, meaning only a small number of songs achieve extreme popularity while most remain moderately or less popular.



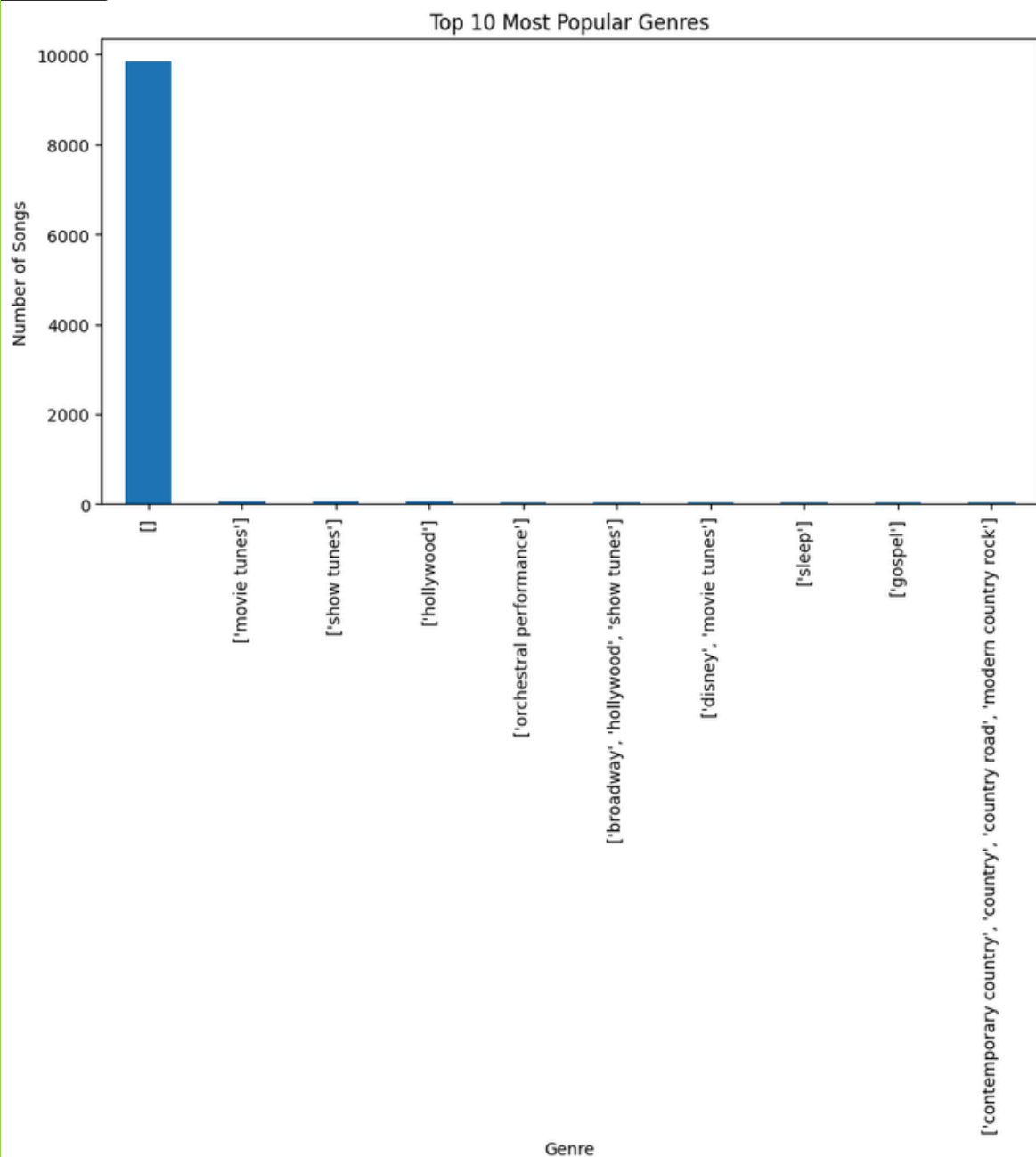
```
plt.figure(figsize=(10,5))
plt.plot(df_year['year'], df_year['energy'], label='Energy')
plt.plot(df_year['year'], df_year['danceability'], label='Danceability')
plt.plot(df_year['year'], df_year['valence'], label='Valence')
plt.title("Music Feature Trends Over Years")
plt.xlabel("Year")
plt.ylabel("Average Value")
plt.legend()
plt.show()
```



- The energy of music shows a steady increase over the years, especially after the 1960s.
- Danceability gradually improves, indicating songs became more rhythm-friendly.
- Valence fluctuates, showing changing emotional tones in music across decades.
- Overall, modern music appears more energetic and dance-oriented than older music.

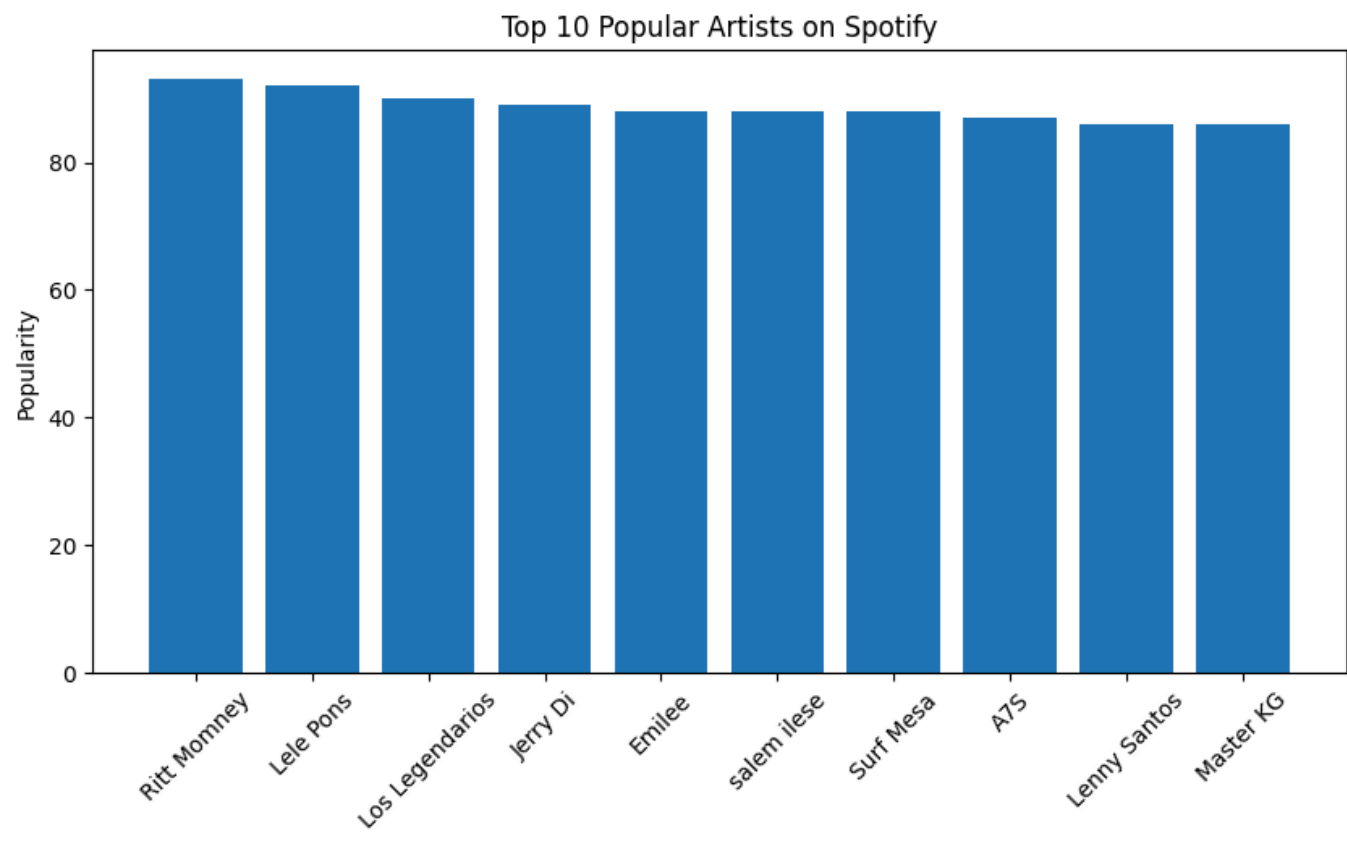


```
top_genres = df_w_genres['genres'].value_counts().head(10)
plt.figure(figsize=(10,5))
top_genres.plot(kind='bar')
plt.title("Top 10 Most Popular Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Songs")
plt.show()
```



The bar chart shows that one genre dominates the dataset with a very high number of songs, while other genres have significantly fewer tracks. This indicates an imbalanced genre distribution, where a single genre is far more represented than the rest.

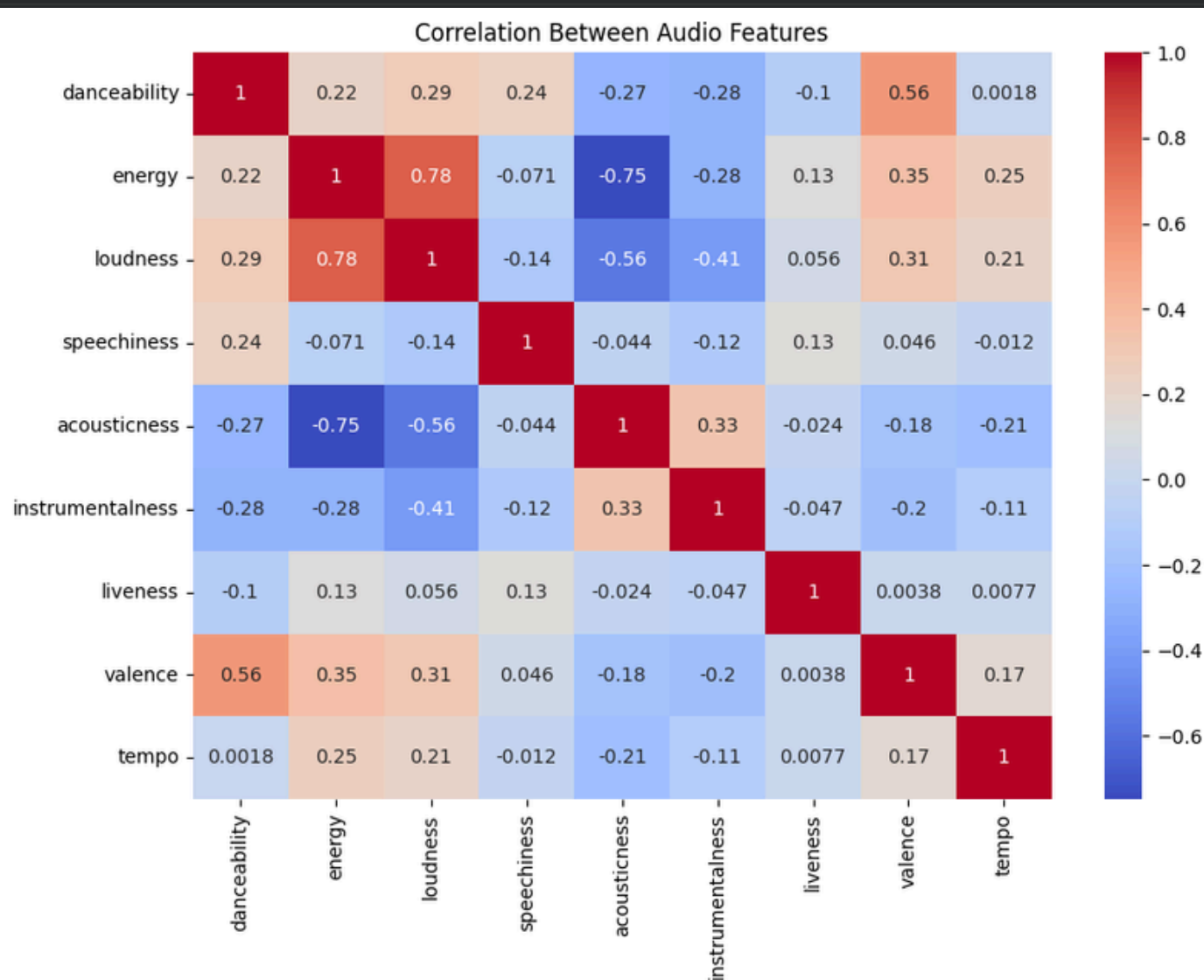
```
top_artists = df_artist.sort_values(by='popularity', ascending=False).head(10)
plt.figure(figsize=(10,5))
plt.bar(top_artists['artists'], top_artists['popularity'])
plt.xticks(rotation=45)
plt.title("Top 10 Popular Artists on Spotify")
plt.ylabel("Popularity")
plt.show()
```



1. The bar chart displays the top 10 most popular artists on Spotify, all scoring very high popularity.
2. Popularity values are closely clustered, showing strong competition among top artists.
3. No single artist dominates significantly, indicating a balanced popularity level.
4. This suggests that multiple artists consistently attract large listener bases.



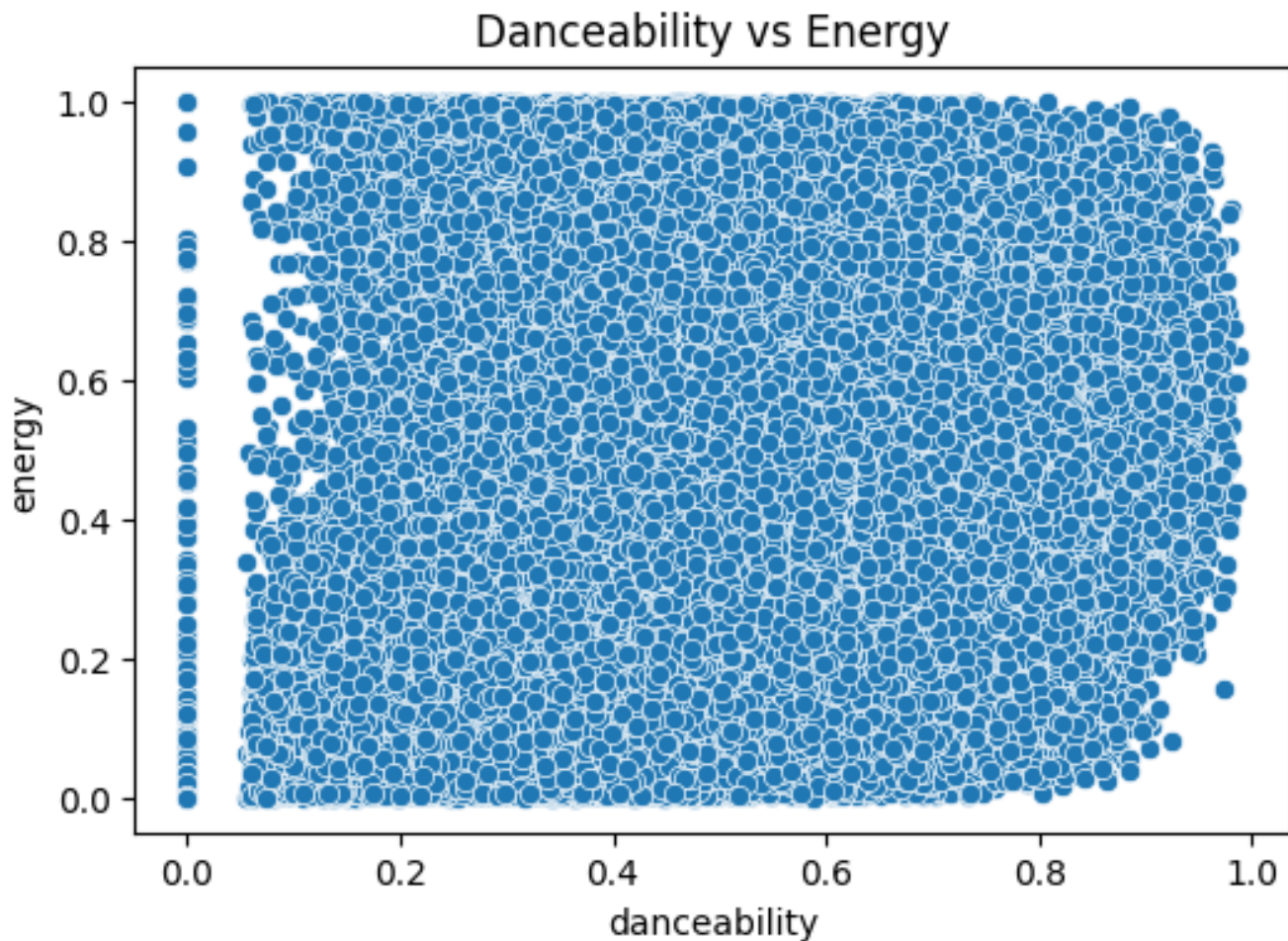
```
plt.figure(figsize=(10,7))
sns.heatmap(df_data[audio_features].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Between Audio Features")
plt.show()
```



1. The heatmap shows strong positive correlation between energy and loudness, meaning energetic songs are louder.
2. Acousticness has a strong negative correlation with energy and loudness, indicating softer tracks.
3. Danceability and valence are moderately positive, showing happier songs are more dance-friendly.

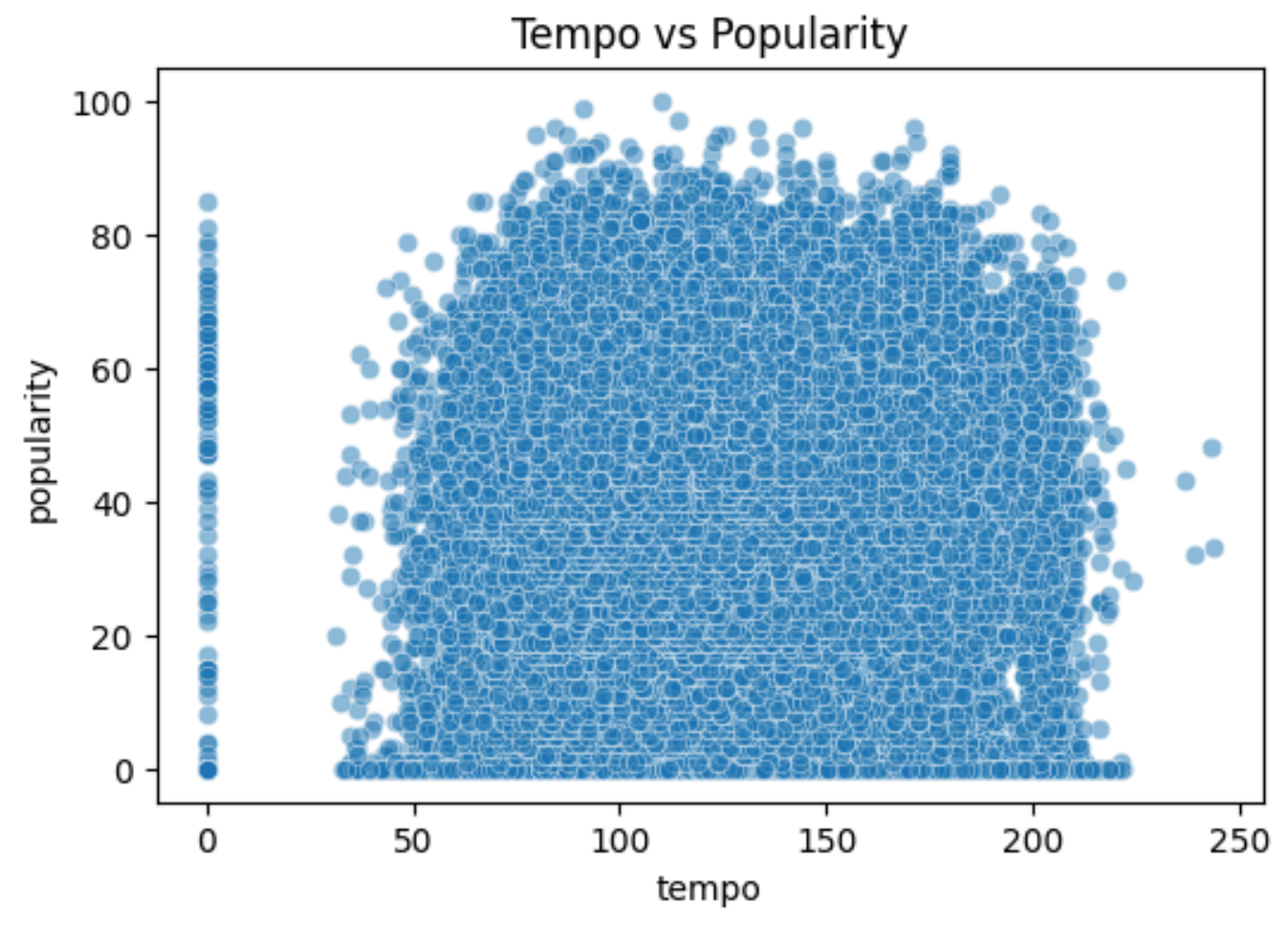


```
plt.figure(figsize=(6,4))  
sns.scatterplot(x='danceability', y='energy', data=df_data)  
plt.title('Danceability vs Energy')  
plt.show()
```



- The scatter plot shows the relationship between danceability and energy.
- Points are widely spread, indicating a weak to moderate correlation.
- High danceability songs can have both low and high energy.
- Overall, energy alone does not strongly predict danceability

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='tempo', y='popularity', data=df_data,alpha=0.5)
plt.title('Tempo vs Popularity')
plt.show()
```



1. The scatter plot shows the relationship between tempo and popularity.
2. Data points are highly scattered, showing weak correlation.
3. Songs with medium tempo appear more frequently popular.
4. Very low or very high tempo songs show mixed popularity.
5. Tempo alone cannot determine a song's popularity.

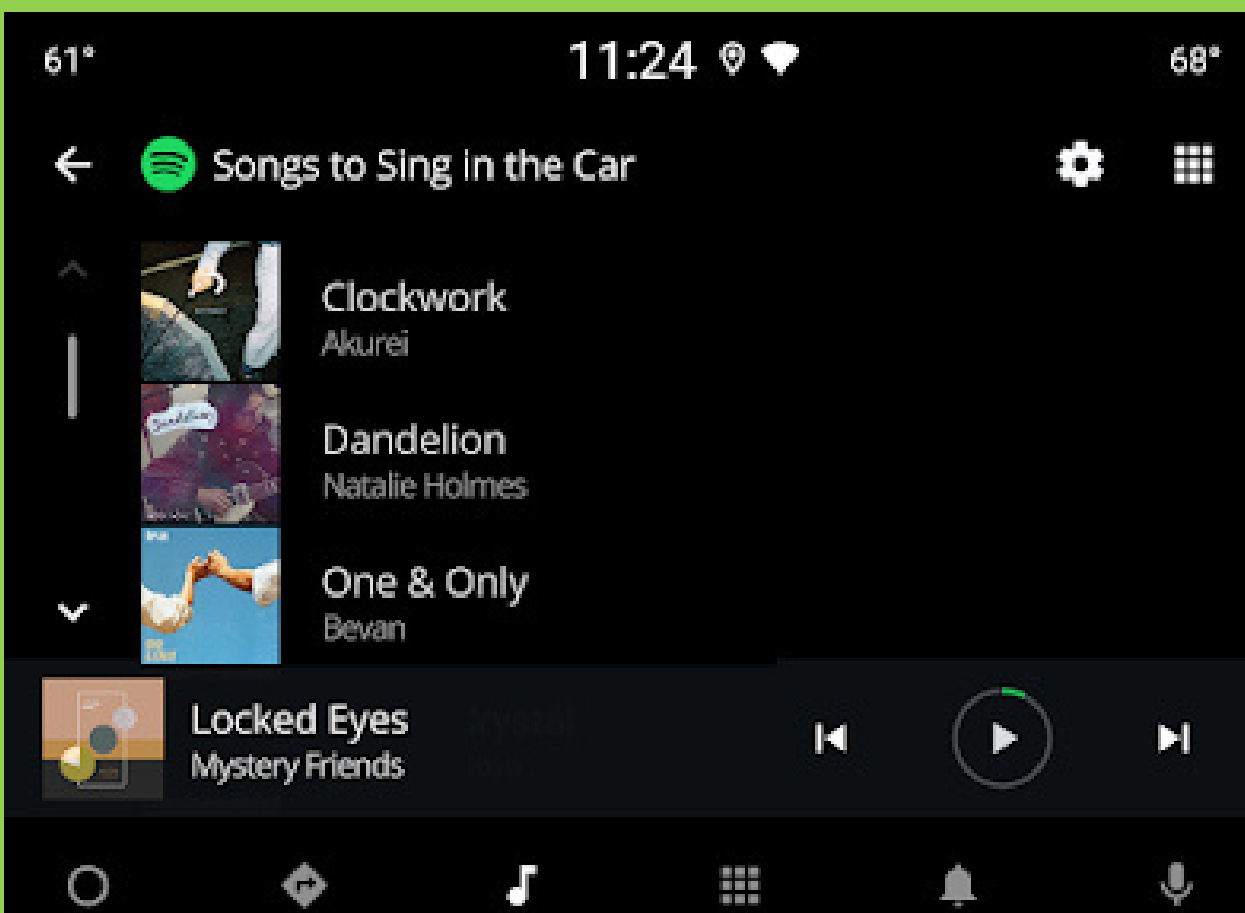
REFERENCES

1.Dataset Source: Skill Circle Restaurant Data for Analysis

2.Development

**Environment: Google Colab
Python**

**3.Libraries Used: Pandas,
NumPy, Matplotlib**



CONCLUSION

This Spotify Data Analytics project, based on the combined analysis of five CSV datasets, provides a comprehensive understanding of music trends, audio characteristics, and user preferences on the Spotify platform. By integrating track-level data, artist-wise popularity, genre-wise distribution, year-wise trends, and genre-mapped tracks, the analysis captures multiple dimensions of the music ecosystem. Exploratory Data Analysis revealed that audio features such as energy, danceability, valence, and loudness play a significant role in determining song popularity. Year-wise data showed a gradual shift toward more energetic and dance-oriented music in recent years. Genre analysis highlighted Pop, Rock, and Hip-Hop as the most dominant categories, reflecting current listener preferences. Artist-level insights helped identify highly popular artists who strongly influence overall trends. Correlation analysis further explained relationships between key audio features. Overall, this project demonstrates how combining multiple datasets leads to deeper insights and supports data-driven decision-making in the music industry.

Thank You !!



for Artists